



UTM
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**SECI1143 – KEBARANGKALIAN STATISTIK & ANALISIS DATA
(PROBABILITY & STATISTICAL DATA ANALYSIS)**

SECTION 03

PROJECT 1:

ADOPTION AND MAINTENANCE OF A HEALTHY LIFESTYLE

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1.0 Introduction

Our topic is “Adoption and Maintenance of a Healthy Lifestyle.” A healthy lifestyle is crucial to maintaining the body's health and preventing diseases. Our research aims to investigate the motivating factors that encourage undergraduate students to adopt healthy habits, understand the challenges they face in maintaining these habits—such as lifestyle changes, external pressures, and lack of motivation—and explore the impact of social media, environmental factors, and personal opinions on their lifestyle choices. Additionally, we aim to evaluate how technology influences their habits and mental health and to analyze their perceptions of what constitutes a healthy lifestyle.

In this project, we will survey undergraduate students at the University of Technology Malaysia (UTM) to gain insights into their methods of adopting and maintaining a healthy lifestyle. We aim to assess their exercise frequency, typical sleep duration, and personal views on the importance of sustaining healthy habits. Through this study, we hope to understand better the lifestyle choices of these students and the factors influencing their health-related decisions.

The questionnaire we prepared, in Google Form, is distributed to undergraduate students of UTM online, through media such as Telegram, WhatsApp, and Instagram and we have successfully gathered 65 responses from the students of UTM. The expectation we have for this research is to investigate the factors that have influenced health-related behavior among students in UTM. These may allow us to have a deeper understanding of how they maintain a healthier lifestyle. We also predict that technology may play an important role in maintaining a healthy lifestyle.

2.0 Data Collection

In this project, we collect the data through surveying via Google Forms. The survey consists of several parts: respondents' personal information, their healthy lifestyles, and factors that motivate respondents to adopt a healthy lifestyle. Some of the questions in Google form will be attached in the appendix part.

No	Question	Answer	Level of Measurement
1	What is your age?	Short answer (in number form)	Nominal
2	What is your gender?	• Male • Female	Nominal
3	What is your faculty?	• Faculty of Computing • Faculty of Mechanical Engineering • Faculty Electrical Engineering • Faculty of Civil Engineering • Faculty of Chemical & Energy Engineering • Faculty of Science • Faculty of Build Environment & Surveying • Faculty of Social Sciences & Humanities • Faculty of Management	Nominal

4	On a scale of 1 to 5, how would you rate your overall satisfaction with your current health?	1(worst) - 5 (best)	Ordinal
5	How often do you exercise?	• Rarely • Occasionally • Sometimes • Often • Always	Ordinal
6	How many times do you exercise in a week?	Short answer	Ratio
7	Do you feel stressed most of the time?	• Strongly disagree • Disagree • Neutral • Agree • Strongly agree	Ordinal
8	How many hours do you usually sleep in a day?	Short answer	Interval
9	Do you think that maintaining a healthy lifestyle is important?	• Yes • No	Nominal
10	What is your main	• Maintaining regular	Nominal

	strategy to sustain a healthy lifestyle over the long term?	routines for exercise • Eating balanced meals • Getting enough sleep • Managing stress effectively • Other	
11	Which factor can motivate you to adopt a healthy lifestyle?	• Peer • Family • Social media • School • Other	Nominal
12	How many fruits do you serve in an average week?	Short answer (let an apple be considered as one unit)	Ratio
13	How much water do you consume in a day?	• Below 500ml • 500ml-1000ml • 1000ml-1500ml • Above 1500ml	Ordinal

3.0 Data Analysis

3.1 Respondent’s Age

Age	Frequency	Percentage
19	6	9.23%
20	34	52.31%
21	16	24.62%
22	2	3.08%
23	2	3.08%
24	1	1.54%
26	3	4.62%
27	1	1.54%
Total	65	100%

Table 1: Frequency Distribution of Respondent’s Age

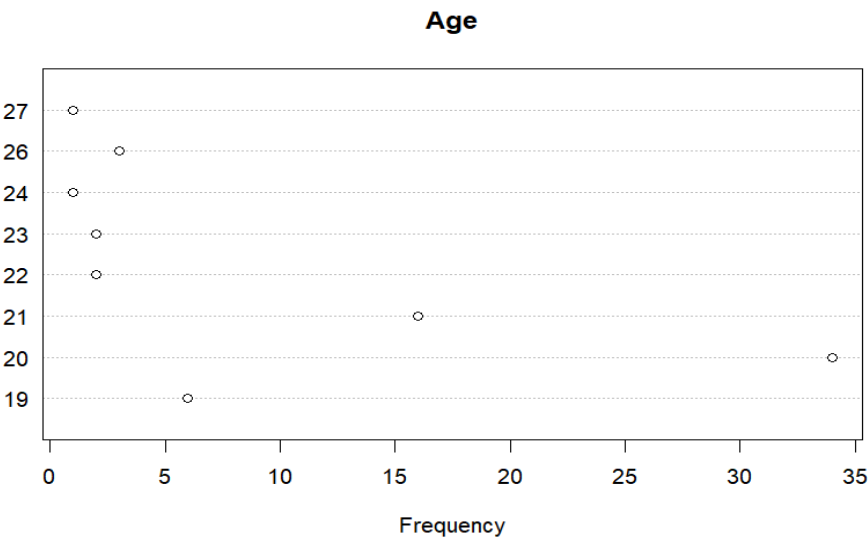


Figure 1: Dot Plot for Age of Respondent

Measure of Central Tendency	Value
Mean	20.75385
Median	20
Mode	20

Table 1.1 Measure of Central Tendency of Respondent's Age

As seen in Table 1 and Figure 1, the age distribution of respondents that have answered the questionnaire is in the range of 19-27. Our research is focused on students from all the faculty in UTM Johor Bahru. Most of the respondents are the age of 20 which has a frequency of 34. There are 16 respondents from the age of 21 and 6 19-year-old respondents. The ages of 22 and 23 with a frequency of 2 and the ages of 24 and 27 hold a frequency of 1. The last is 26 years old with a frequency of 3. Additionally, the mean, mode and median can be seen in Table 1.1, in which the mode is 20, the median is 20, and the mean is 20.75385.

3.2 Respondent's Gender

Gender	Frequency	Percentage
Female	34	52.31%
Male	31	47.69%
Total	65	100%

Table 2: Frequency Distribution of Respondent's Gender

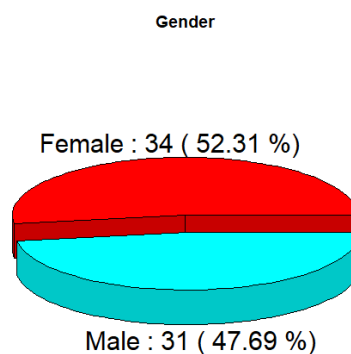


Figure 2: Pie Chart for Respondent's Gender

From Table 2 and Figure, which is the frequency table and pie chart, we know that the frequency of females is 34 and males is 31. Thus, we can know the percentage of females is 52.3% and males is 47.7% and from this percentage, we can know the female respondents are more than male respondents. Additionally, the red color in the pie chart represents females and blue represents males.

3.3 Faculty for Respondent

Faculty	Frequency	Percentage
Faculty Electrical Engineering	10	15.38%
Faculty of Build Environment & Surveying	6	9.23%
Faculty of Chemical & Energy Engineering	7	10.77%
Faculty of Civil Engineering	6	9.23%
Faculty of Computing	18	27.69%
Faculty of Management	5	7.69%
Faculty of Mechanical Engineering	5	7.69%
Faculty of Science	3	4.62%
Faculty of Social Sciences & Humanities	5	7.69%
Total	65	100%

Table 3: Frequency Distribution of Faculty of Respondent

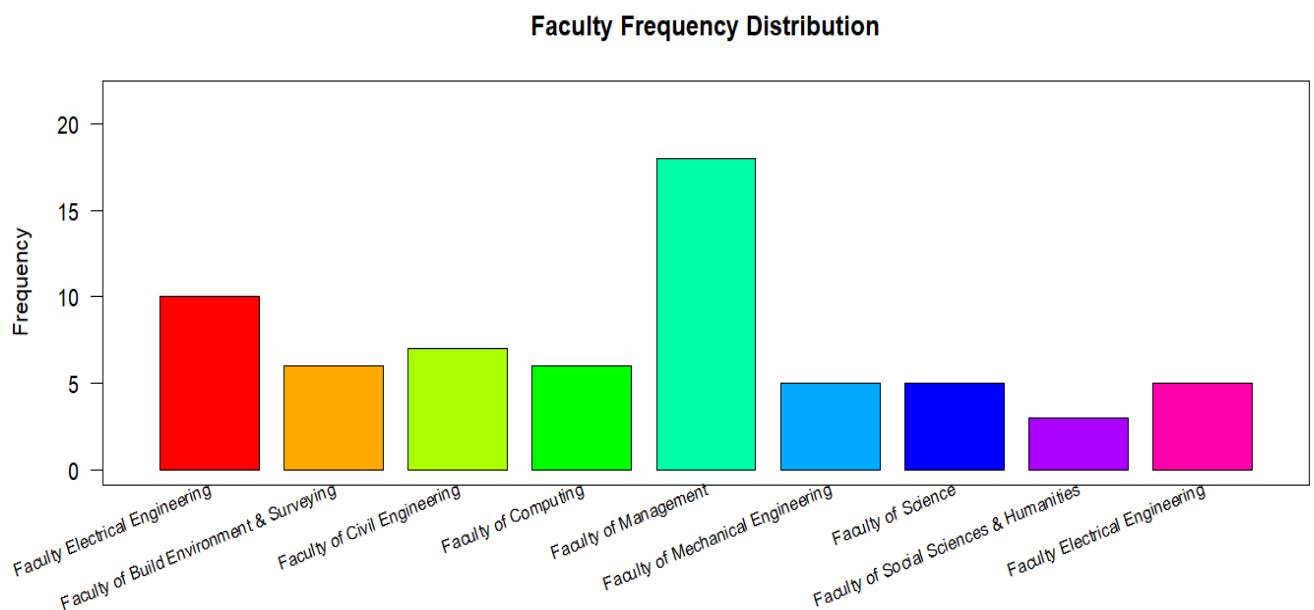


Figure 3: Bar Chart for Respondent's Gender

As indicated in Table 3 and Figure 3, the 65 respondents have come from 9 different faculties in UTM Johor Bahru. First, we have 18 respondents from the Faculty of Computing and 10 from the Faculty of Electrical Engineering. There are 7 respondents from the Faculty of Chemical & Energy Engineering. The Faculty of Build Environment & Surveying and the Faculty of Civil Engineering have 6 respondents. We have three faculty that have a frequency of 5, the Faculty of Management, the Faculty of Mechanical Engineering and the Faculty of Social Science & Humanities. The last is the Faculty of Science which has a frequency of 3.

3.4 Respondents rate their overall satisfaction with their current health.

Scale	Frequency	Percentage
1	1	1.54%
2	4	6.15%
3	18	27.69%
4	31	47.69%
5	11	16.92%
Total	65	100%

Table 4: Frequency Distribution of respondents rate of their overall satisfaction with their current health

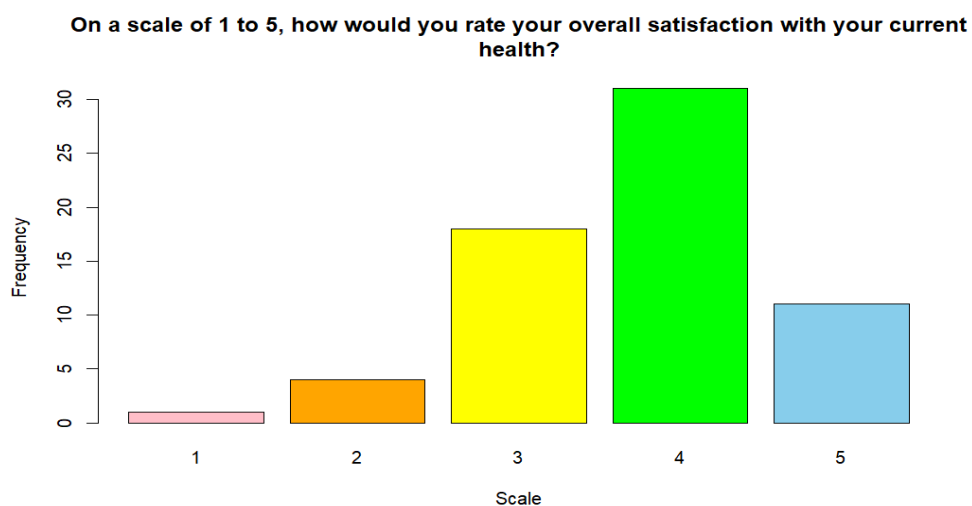


Figure 4: Bar Chart for the respondents rate of their overall satisfaction with their current health

Table 4 and Figure 4 show the opinions of respondents on their overall satisfaction with their current health. Only 1 respondent chose a scale of 1, 4 respondents chose a scale of 2, 18 respondents chose a scale of 3, 31 respondents chose a scale of 4 and 11 respondents chose a scale of 5. We know that the majority of respondents are satisfied with their health. This is because they have a healthy lifestyle by eating a balanced diet and doing exercise regularly.

3.5 Respondents' exercise frequency

Scale	Frequency	Percentage
Rarely	20	30.80%
Occasionally	9	13.80%
Sometimes	22	33.80%
Often	11	16.90%
Always	3	4.60%
Total	65	100%

Table 5: Scale and Percentage of Frequency for How Often Respondents Exercise

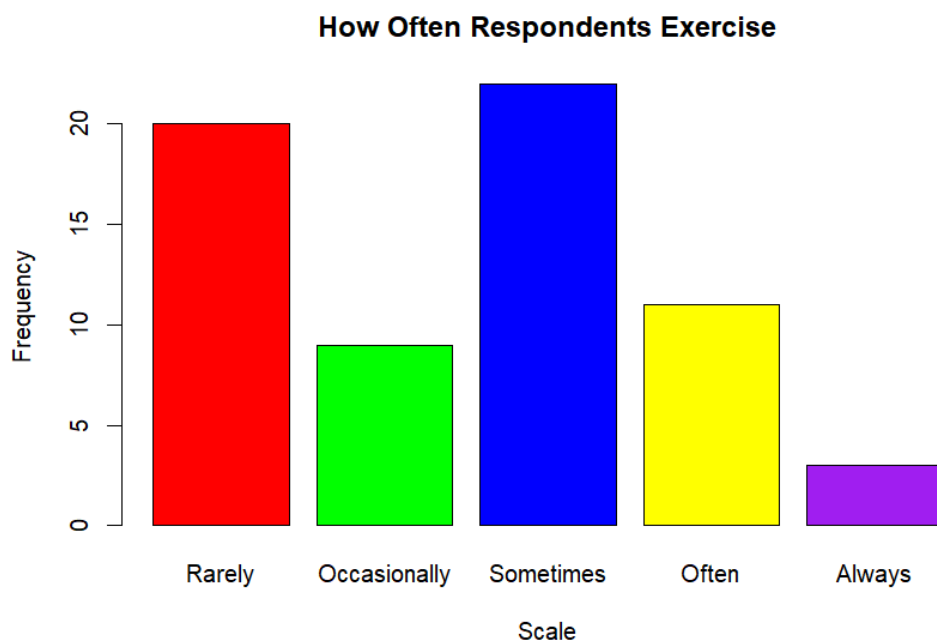


Figure 5: Bar chart of Frequency for How Often Respondents Exercise

According to Table 5 and Figure 5, provide an overview of 65 respondents about their exercise frequency, ranging from "Rarely" to "Always". 22 respondents exercise sometimes, making it the largest category, they do exercise regularly but not consistently. Next, followed by 20 respondents who rarely exercise, 11 respondents who often maintain a stable routine, 9 respondents who do so occasionally, and 3 respondents who always incorporate exercise into their daily lives, representing the smallest segment, these participants fully integrate exercise into their daily routines.

The bar chart effectively highlights these distributions, with the tallest bars for "Sometimes" and "Rarely" indicating that while a large number of respondents engage in exercise, it is not a frequent or consistent habit for most.

The significant number of respondents who rarely or only occasionally exercise suggests a need for targeted health promotion efforts to foster more consistent physical activity, which could greatly enhance public health outcomes.

3.6 Respondents' weekly exercise frequency

Times (in a week)	Frequency
0	12
1	21
2	21
3	7
4	1
5	3
Total	65

Table 6: Frequency of Weekly Exercise Sessions Among Respondents

Measures of Central Tendency	Value
Mean	1.584615
Median	1
Mode	1

Table 6.1 Measures of Central Tendency of Respondents' weekly exercise frequency

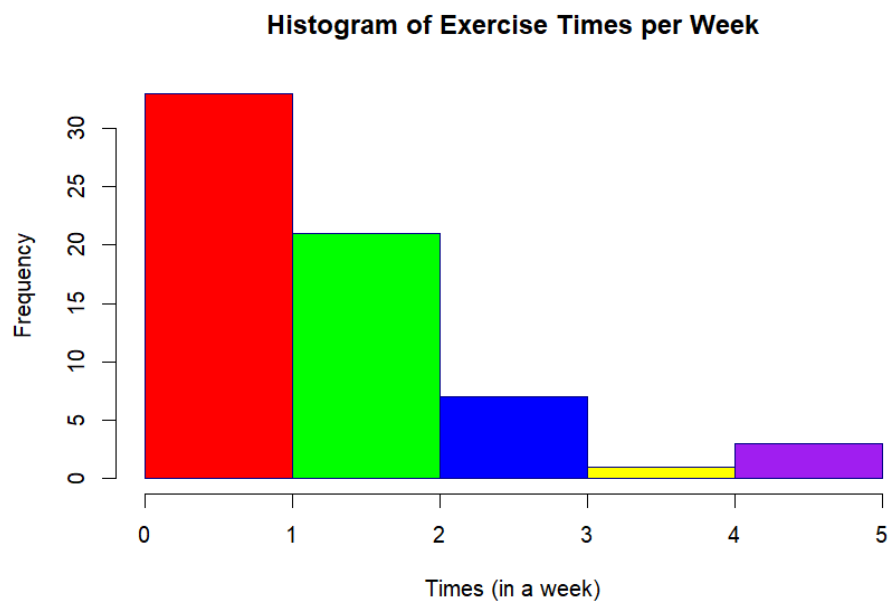


Figure 6: Histogram of Frequency of Weekly Exercise Sessions Among Respondents

From Table 6 and Figure 6, we can see the frequency of undergraduates doing exercise in a week, it is distributed from 0 to 5 times. Among the 65 respondents surveyed, the majority of respondents exercise between zero and two times per week. Specifically, 21 respondents each reported exercising once and twice a week, while 12 respondents reported not exercising at all. This shows a tendency towards minimal physical activity among the majority.

There is a clear decline in frequency as the number of exercise sessions increases. Only 7 respondents exercise three times a week, one exercises four times, and three manage to

exercise five times a week. This steep drop-off indicates that maintaining a more regular exercise regimen is less common.

The histogram vividly depicts this declining trend with tall bars for lower frequencies (0-2 times a week) and much shorter bars for higher frequencies (3-5 times a week), visually emphasizing the rarity of frequent exercise among the participants.

The drastic drop in frequency for those exercising three times or more indicates that maintaining a more intensive exercise schedule might be challenging for many, possibly due to time constraints or lack of motivation. The data underscores the need for initiatives to promote more frequent physical activity, considering the benefits of regular exercise in managing stress, improving mental health, and enhancing physical well-being.

Additionally, it can be seen from Table 6.1 that the data we gathered will mostly be from doing exercise 1-2 times a week, as our median and mode both are 1, and the mean being 1.584615.

3.7 Do you feel stressed most of the time?

Likert Scale	Frequency	Percentage
Strongly Agree	2	3.08%
Agree	18	27.69%
Neutral	27	41.54%
Disagree	11	16.92%
Strongly Disagree	7	10.77%
Total	65	100%

Table 7: Likert Scale Frequency of whether Respondents Feeling Stressed Most of the Time

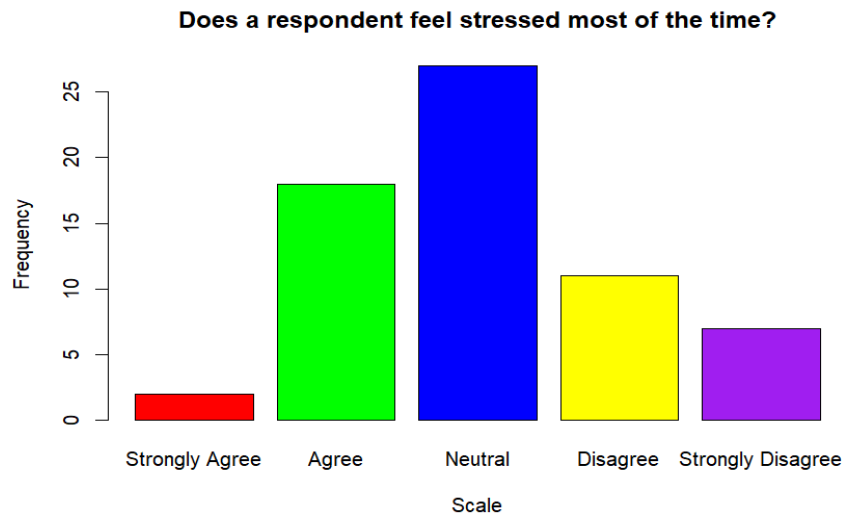


Figure 7: Bar Chart of Likert Scale Frequency of whether Respondents Feeling Stressed Most of the Time

It is about whether respondents feel stressed most of the time, as shown in Table 7 and Figure 7. In this bar graph, it shows that the x-axis is the agreement level and the y-axis is the frequency of the respondents that we collected.

The analysis of Table 7 and its corresponding bar chart, which surveyed 65 respondents about their stress levels, shows a diverse range of experiences with stress among UTM students. Most respondents do not feel overwhelmingly stressed, with the largest group (27 respondents) selecting "Neutral," indicating variable or moderate stress levels. Meanwhile, 20 respondents expressed that they do feel stressed most of the time ("Strongly Agree" and "Agree"), highlighting a significant portion that may require more effective stress management resources or support systems. On the other end, 18 respondents ("Disagree" and "Strongly Disagree") feel little to no frequent stress, suggesting good coping mechanisms or a naturally lower stress level in their lives.

This mixed response underlines the need for personalized approaches to stress management in educational settings, ensuring that interventions can address the varying needs of students, helping improve their mental health and academic success.

3.8 Respondents' sleeping time in a day

Time (hours)	Frequency
2	1
4	2
5	11
6	16
7	24
7.5	1
8	9
9	1
Total	65

Table 8: Frequency of sleeping time in a day among respondents

Measures of Central Tendency	Value
Mean	6.423077
Mode	7
Median	7

Table 8.1: Measures of central tendency of sleeping time in a day among respondents

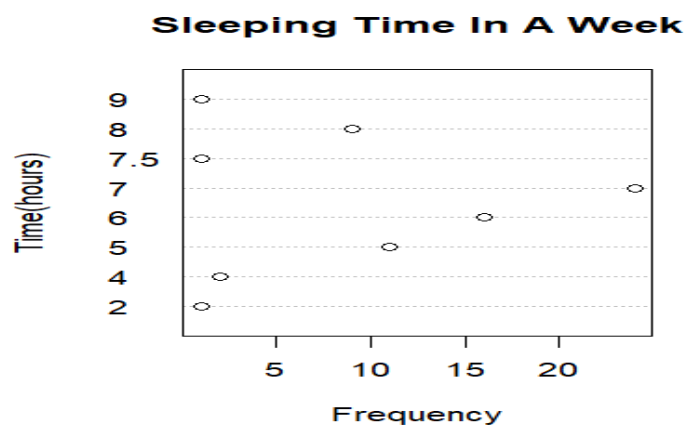


Figure 8: Dot plot of sleeping time in a day among respondents

According to Table 8 and Figure 8, they show that most of the respondents have an average sleeping time of 6 hours and 7 hours in a day. In addition, 10 respondents sleep more than 7 hours a day. This is a good phenomenon to be seen since most young people nowadays sleep less than recommended. However, there are 14 respondents who sleep less than 6 hours due to several factors like being busy dealing with their studies and co-curricular activities. If this situation lasts for a long time, it will definitely have negative effects on the respondents' mental and physical health.

3.9 Respondents' opinion on the importance of maintaining a healthy lifestyle

Answer	Frequency	Percentage (%)
Yes	65	100
No	0	0
Total	65	100

Table 9: Frequency of respondents' answer on the importance of maintaining a healthy lifestyle

Opinion Of The Respondents On The Importance Of Maintaining A Healthy Lifestyle

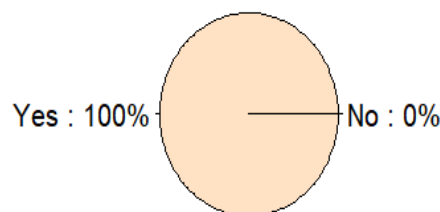


Figure 9: Pie chart of respondents' opinion on the importance of maintaining a healthy lifestyle

Based on Table 9 and Figure 9, they illustrate that all the 65 respondents think that maintaining a healthy lifestyle is important in our daily routine. This is definitely what the result we expected before as we hope that everyone has the awareness of adopting a healthy lifestyle.

3.10 Respondents' main strategy to sustain a healthy lifestyle over the long term

Strategies	Frequency	Percentage (%)
Maintaining regular routines for exercise	8	12.31
Eating balanced meals	12	18.46
Getting enough sleep	23	35.38
Managing stress effectively	20	30.77
Others	2	3.08
Total	65	100

Table 10: Frequency of respondents' opinion on the main strategy to sustain a healthy lifestyle over the long term

Main strategy to sustain a healthy lifestyle over long term

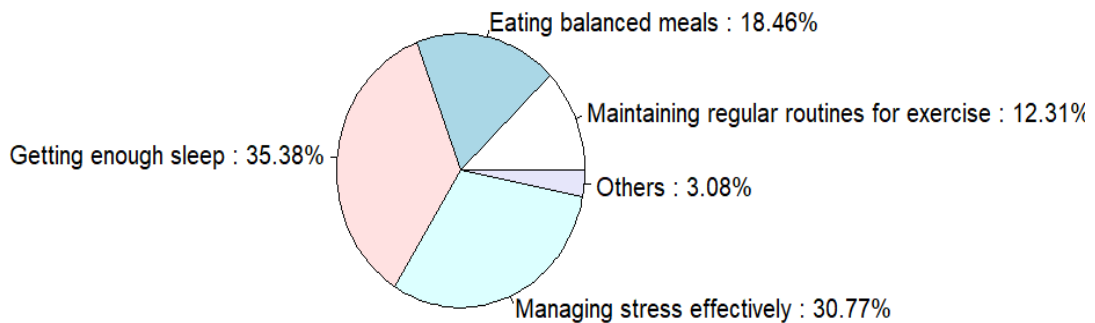


Figure 10: Pie chart of respondents' opinion on the main strategy to sustain a healthy lifestyle over the long term

Table 10 and Figure 10 clearly show how 65 respondents adopted a healthy lifestyle by using their own main strategy. 8 respondents expressed their opinion by saying that they keep maintaining a regular routine for exercise. At the same time, 12 respondents choose the option of eating balanced meals, 23 respondents choose the option of getting enough sleep and 20 respondents choose the option of managing stress effectively. There is a respondent having another strategy to sustain a healthy lifestyle over the long term which is being away from social media. There is 1 more respondent giving his or her option that he or she does not have a main strategy to sustain a healthy lifestyle.

3.11 The primary driver of respondents' adoption of a healthy lifestyle

Which factor can motivate you to adopt a healthy lifestyle?	Frequency	Probability
Family	27	41.54
Peer	18	27.69
Social media	3	18.46
School	12	4.62
Others	5	7.69

Table 11: Frequency of respondents' opinions regarding what factors motivate them to adopt a healthy lifestyle

Factors Motivating UTM Students to Adopt a Healthy Lifestyle

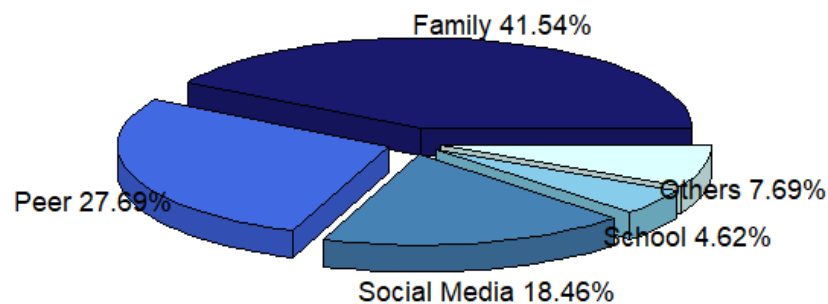


Figure 11: Pie chart of factors motivating respondents to adopt a Healthy Lifestyle

According to Table 11 and the pie chart,, the provided data offers insights into the factors influencing respondents' adoption of a healthy lifestyle. Family emerges as the most motivational factor, with a frequency of 27 over 65, constituting approximately 41.54% of responses. Following closely behind is peer influence, with 18 responses, comprising 18.46%. This highlights the vital role of familial and peer support in motivating students to embrace healthy behaviors. Surprisingly, social media appears to have a comparatively lower impact, with only 12 responses, representing around 18.46%. Similarly, school-based influences also seem less significant, with a frequency of 3, making up only 4.62% of responses. Additionally, the "Others" category contributes 5 responses, which is 7.69%, showing the presence of additional motivators beyond those explicitly mentioned. These findings underscore the significance and amazingness of social circles in shaping lifestyle choices.

3.12 The average amount of fruits respondents serve per week

How many fruits do you serve in an average week ?	Frequency
0	12
1	13
2	13
3	13
4	9
5	1
6	1
7	2
16	1

Table 12.1 : Average fruit serving frequency of respondents

Measures of Central Tendency	Value
Mode	1,2,3
Mean	2.375
Median	2

Table 12.2 : Average fruit serving frequency of respondents

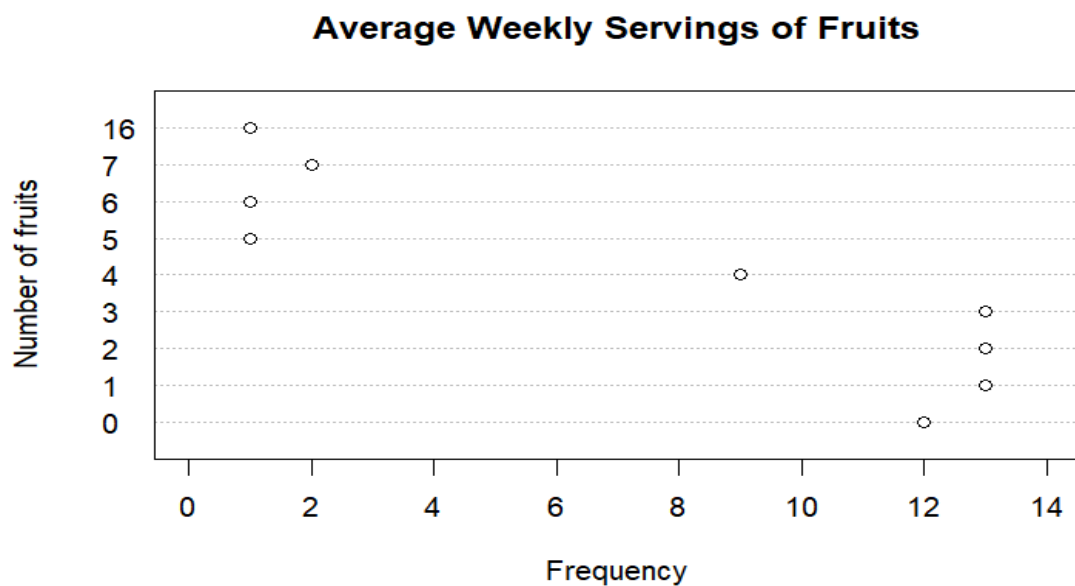


Figure 12: Dot plot regarding average fruits served by respondents per week

Based on Table 12.1 and the dot plot, the data depicts the frequency distribution of the average amount of fruits served per week by respondents. Examining the plot, we can observe that the visual results reveal a right-skewed distribution, with 0 to 4 fruits per week, which means the majority of respondents serve a lower frequency of fruits in their weekly diet. Conversely, fewer respondents reported serving higher quantities of fruits per week, as evidenced by only two respondents out of 65 indicating they consume seven fruits per week, while only one consuming more than seven. Referring to Table 12.2, we can perceive that the modes are 1, 2, and 3, which all depict lower fruit counts, and three of them have a frequency of 13, closely followed by 0 having a frequency of 12. The frequency is extremely discrepant between the lower range (0 to 4) and the higher range (5 to 16). Therefore, our mean obviously tends to the lower range, which is 2.375, and the median is 2. Hence, it is imperative for UTM students to increase their fruit intake as part of adopting a healthy lifestyle, ultimately contributing to improved physical well-being.

3.13 Respondents' water drinking in a day

Daily Water Consumption (ml)	Frequency	Percentage (%)
Below 500	2	3.08
500-1000	21	32.31
1000-1500	24	36.92
Above 1500	18	27.69

Table 13: Frequency of respondents' drinking of water per day

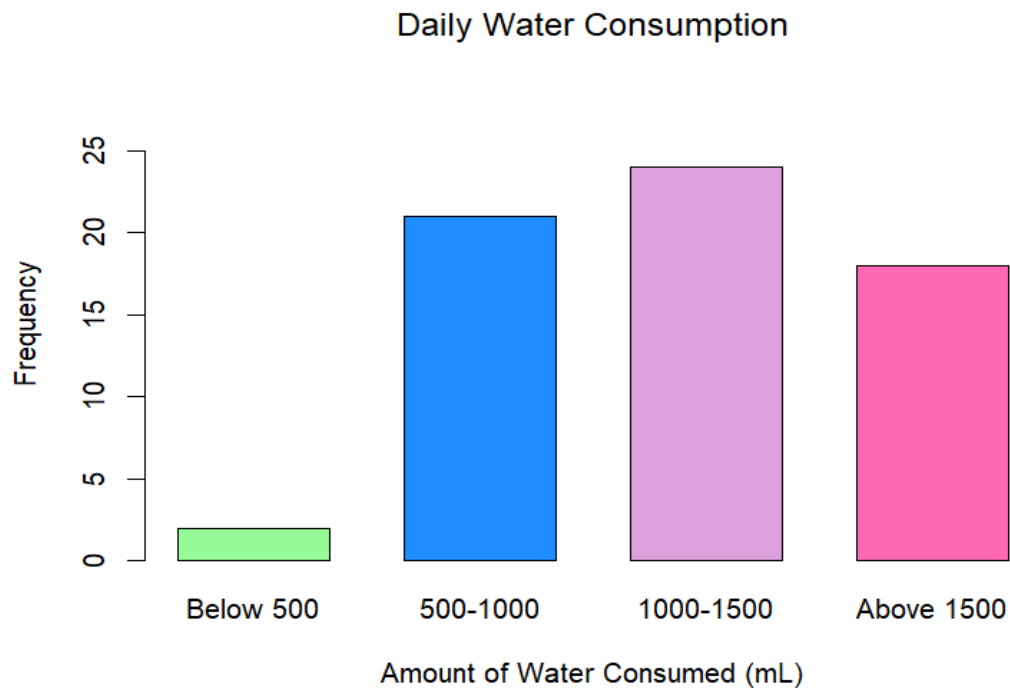


Figure 13: Bar chart of respondents' daily amounts of water consumption

The data in Table 13 and the bar chart reveal the respondents' daily water intake. The bar chart clearly shows four different categories of water consumption in mL. Notably, the fewest respondents reported consuming water below 500 mL, with a frequency of 4, while the rest of the three categories have an average frequency. The larger proportion reported intake within the 1000–1500 mL range, directly after 500–1000 mL, and lastly above 1500 mL, respectively, 24, 21, and 18 responses out of 65. However, an adult should drink more than 2000 mL per day, which is approximately 8 cups of water. Thus, every respondent should increase the amount of their daily water consumption to maintain their body's functions.

4.0 Conclusion

In conclusion, our study which involved 60 respondents focused on investigating the adoption and maintenance of a healthy lifestyle among students at UTM. Through a structured survey, data was collected on various aspects. From the collected and analyzed data, we can see that most of the respondents are in the ages of 20 and 21. We come out with the measure of the central tendency of respondents' age with mean equal to 20.75385, median equals to 20 and mode equals to 20 as well. Furthermore, there are 34 females and 31 males over 65 respondents. Additionally, most of the respondents are satisfied with their current health. However, based on the analyzed data regarding respondents' exercise frequency, there are less number of respondents who have a habit of exercising regularly, 30.80% of respondents rarely exercise and only 4.60% of respondents always exercise. From the table of measures of central tendency of respondents' weekly exercise frequency, we obtain the mean which is 1.584615, and 1 for both median and mode. In the aspect of feeling stressed most of the time, 41.54% of the respondents feel neutral about it. Moreover, most of the respondents have more than 6 hours of sleep. By calculating the measures of central tendency regarding this aspect, we get the mean value equals 6.423077, and 7 for both mode and median. After that, it is good to see that all the 65 respondents think that maintaining a healthy lifestyle is important. Besides, most of the respondents' main strategy to sustain a healthy lifestyle over the long term is to get enough sleep, the percentage is 35.38% over 100.00%. After that, 41.54% of the respondents choose family as their primary driver in adopting a healthy lifestyle. It is surprising that 12 respondents do not consume any fruits per week on average. In this aspect, we come out with the measures of central tendency with 3 mode values which are 1, 2 and 3, median value which is 2.375 and median with the value of 2. For the last aspect about daily water

consumption, 3.08% of the respondents consume less than 500 ml of water per day while 27.69% of respondents consume more than 1500 ml of water per day.

Mathematically, we have learned to utilize different types of data measurements, including nominal, ordinal, interval, and ratio scales, to analyze and interpret the responses. The study has utilized statistical analysis, involving calculations and algorithms to interpret the gathered information. Moreover, technology played a crucial role in data collection, analysis, and potentially in influencing participants' habits through the impact of social media and other technological advancements. From a technological standpoint, we have learned to use digital tools or software which is R programming for data collection, analysis, and presentation. By using this software, we are able to create several types of graphs or charts easily by applying a few lines of codes.

Lastly, this study provides valuable insights into the factors influencing healthy lifestyle choices among university students and highlights the importance of technology in shaping health behaviors in today's digital age.

5.0 Appendix

Screenshot of Google Form

QuestionsResponses65Settings



Section 1 of 2

Adoption and Maintenance of a Healthy Lifestyle

Hiii!! Thank you for taking the time to complete our survey, and welcome!

We are students from Faculty of Computing, Universiti Teknologi Malaysia.

The purpose of this form is to **gather information on how students of UTM adopt and maintain healthy lifestyles**. The insights you provide will help us understand the challenges and success factors associated with living healthily. We are interested in a variety of factors including dietary habits, physical activity, mental health, and social dynamics.

Your responses will remain confidential and be used solely for the purpose of this study. It would just take a few minutes to complete the form. Please answer each question as honestly as possible. We appreciate your valuable insights and time. **Thank you for helping us explore this important topic!**

Gender *

☐ Male

☐ Female

What is your age *

Short-answer text

What is your faculty *

1. Faculty of Computing
2. Faculty of Mechanical Engineering
3. Faculty Electrical Engineering
4. Faculty of Civil Engineering
5. Faculty of Chemical & Energy Engineering
6. Faculty of Science
7. Faculty of Build Environment & Surveying
8. Faculty of Social Sciences & Humanities
9. Faculty of Management

On a scale of 1 to 5, how would you rate your overall satisfaction with your current health? *

worst 1 2 3 4 5 best

☐ ☐ ☐ ☐ ☐

How often do you exercise? *

- ☐ Rarely
- ☐ Occasionally
- ☐ Sometimes
- ☐ Often
- ☐ Always

How many times do you exercise in a week? *

Short-answer text

Do you feel stressed most of the time? *

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

How many hours do you usually sleep? *

Short-answer text

Do you think that maintaining a healthy lifestyle is important? *

- ☐ Yes
- ☐ No

What is your main strategy to sustain a healthy lifestyle over the long term? *

- ☐ Maintaining regular routines for exercise
- ☐ Eating balanced meals
- ☐ Getting enough sleep
- ☐ Managing stress effectively
- ☐ Other...



Which factor can motivate you to adopt a healthy lifestyle? *

- ☐ Peer
- ☐ Family
- ☐ Social media
- ☐ School
- ☐ Other...

How many fruits do you serve in an average week (let an apple considered as one unit) *

Short-answer text

How much water do you consume in a day? *

- ☐ Below 500ml
- ☐ 500ml-1000ml
- ☐ 1000ml-1500ml
- ☐ Above 1500ml

We appreciate your participation in this Google Form. Thank you for your assistance!



Description (optional)

Image title

